

JAVA INTERFACE PROGRAM

Application: Online Payment | Program 4

1. Aim

To write a Java program in which one class implements two interfaces, Payment and Receipt, to process an online payment and generate a payment receipt.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program4.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create the Payment interface with `pay()`.
6. Create the Receipt interface with `generateReceipt()`.
7. Create the OnlinePayment class implementing both interfaces.
8. Create an OnlinePayment object and call `pay()` and `generateReceipt()`.

3. Algorithm

1. Start the program.
2. Create the Payment interface with `pay()`.
3. Create the Receipt interface with `generateReceipt()`.
4. Create OnlinePayment implementing Payment and Receipt.
5. Declare amount as 2500 and discount as 200.
6. Calculate final amount as amount – discount.
7. Display amount, discount and final amount.
8. Generate the payment receipt.
9. Create an OnlinePayment object in `main()`.
10. Call `pay()` and `generateReceipt()`.
11. Stop the program.

4. Explanation of the Program

Payment interface: It declares the `pay()` method for processing payment details.

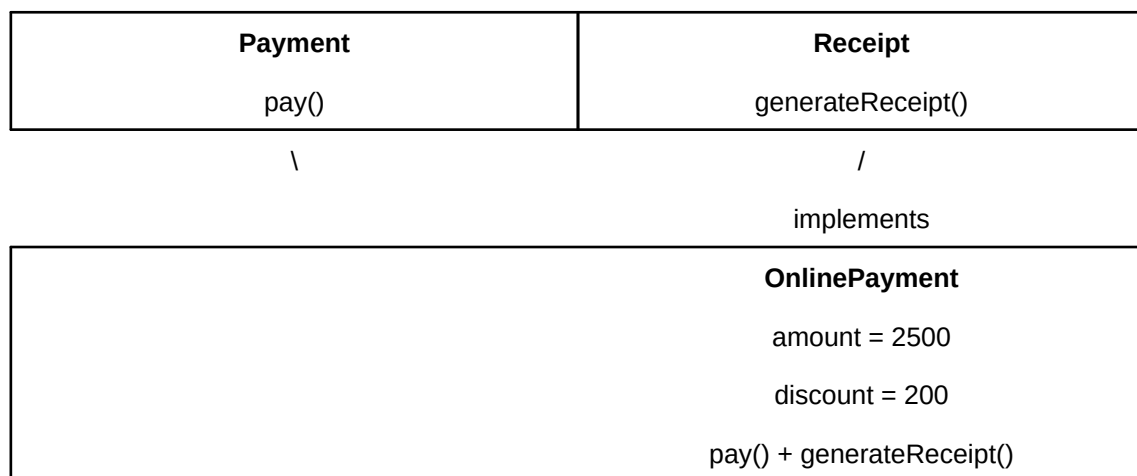
Receipt interface: It declares the `generateReceipt()` method for generating the payment receipt.

OnlinePayment class: It implements both Payment and Receipt interfaces and provides implementations of their methods.

Payment calculation: The final amount is calculated by subtracting the discount from the original amount.

Multiple interface implementation: The class OnlinePayment uses implements Payment, Receipt, allowing one class to implement methods from two interfaces.

5. Interface Implementation Diagram



6. Java Source Code

```
// Program 4: C1 implements I1, I2
// Application: Online Payment

interface Payment {
    void pay();
}

interface Receipt {
    void generateReceipt();
}

class OnlinePayment implements Payment, Receipt {

    double amount = 2500;
    double discount = 200;

    public void pay() {
        System.out.println("Amount = " + amount);
        System.out.println("Discount = " + discount);
        System.out.println("Final Amount = " + (amount - discount));
    }

    public void generateReceipt() {
        System.out.println("Payment receipt generated");
    }
}
```

```
    }  
}  
  
public class Program4 {  
    public static void main(String[] args) {  
  
        OnlinePayment p = new OnlinePayment();  
  
        p.pay();  
        p.generateReceipt();  
    }  
}
```

7. Output of the Program

```
Amount = 2500.0  
Discount = 200.0  
Final Amount = 2300.0  
Payment receipt generated
```

8. Result

Thus, the Java program successfully demonstrates multiple interface implementation by allowing the OnlinePayment class to implement both Payment and Receipt, calculate the final payment amount, and generate the payment receipt.